

Problem Set for 03 April 2025

Definition (Eigenvector, Eigenvalue, and Eigenpair) Let $f : V \rightarrow V$ be an endomorphism. If and only if there exists

$$f(v) = \lambda v, \quad (\lambda \in \mathbb{F}, \quad v \neq 0),$$

we say:

- v is an eigenvector of f ;
- λ is an eigenvalue of f ; and
- (λ, v) is an eigenpair.

Exercise 1 Determine all eigenvalues of the linear endomorphism:

$$\varphi : \mathbb{R}[x] \rightarrow \mathbb{R}[x], \quad f(x) \mapsto \int_0^x f(t) \, dt.$$

- Hint: the expression might be misleading, set $V := \mathbb{R}[x]$, one has

$$\varphi : V \rightarrow V, \quad f \mapsto \begin{bmatrix} \mathbb{R} & \rightarrow & \mathbb{R} \\ x & \mapsto & \int_0^x f(t) \, dt \end{bmatrix}.$$

Exercise 2 Given $a, b \in \mathbb{R}$, determine all eigenpairs of the linear endomorphism:

$$\varphi : \mathbb{R}[x] \rightarrow \mathbb{R}[x], \quad f(x) \mapsto f(ax + b).$$

- Hint: find fixed point of $ax_0 + b = x_0$.

Exercise 3 Let V the space of smooth, bounded, and real valued functions over $\mathbb{R}$, which means that $f \in V$ maps $\mathbb{R}$ to some $[a, b] \subseteq \mathbb{R}$, and $\frac{d^{\forall k \geq 1}}{dx^k} f(x)$ exists. Determine all eigenpairs of the linear endomorphism:

$$\varphi_k : V \rightarrow V, \quad f(x) \mapsto \frac{d^k}{dx^k} f(x).$$

- Hint: for $P \in \mathbb{Z}[x]$ with distinct roots $\{z_k\}_{k=1}^{\deg n} \subseteq \mathbb{C}$, the solution of $P(\frac{d}{dx})(f) = 0$ forms a linear space spanned by $\{e^{z_i x}\}_{i=1}^{\deg n}$. For instance, $f'' - 3f' + 2f = 0$ has a linear space of the solutions $C_1 e^x + C_2 e^{2x}$ for $C_i \in \mathbb{R}$.
- **(Optional)** How would you like to define $\frac{d^{1/2}}{dx^{1/2}} f(x)$?

Exercise 4 Let $f : V \rightarrow V$ be an $\mathbb{F}$ -linear endomorphism. Prove or disprove the following statement:

- v is an eigenvector of f , if and only if $v \in \ker(\lambda \cdot \text{id} - f)$.

Exercise 5 (Optional) Let $V = C^0([0, 1])$, i.e., the space of continuous real-valued functions over the closed interval $[0, 1]$. Determine all eigenpairs of the operator:

$$V \rightarrow V, \quad f(x) \mapsto \int_0^1 K(x, y) \, dy,$$

where the kernel function is defined as:

$$K(x, y) := \min(x, y) - x \cdot y.$$

- Hint: Under appropriate conditions,

$$\frac{d}{dx} \int_{f(x)}^{g(x)} H(x, y) \, dy = H(x, g(x)) - H(x, f(x)) + \int_{f(x)}^{g(x)} H_x(x, y) \, dy.$$

Exercise 6 (optional) Let V be the space of absolutely integrable functions from $\mathbb{R}$ to $\mathbb{C}$. Define the Fourier transform:

$$\mathcal{F} : V \rightarrow V, \quad f(x) \mapsto \widehat{f}(u) := \frac{1}{2\pi} \int_{-\infty}^{+\infty} f(t) e^{-itu} \, dt.$$

1. Is $\mathcal{F}$ a well-defined $\mathbb{C}$ -linear endomorphism? If unsure, consult Professor Tao for appropriate spaces V , or simply take

$$W := \bigcap_{k \geq 0} V_k, \quad V_0 := V, \quad V_{k+1} := \mathcal{F}(V_k) \cap V_k$$

which ensures that $\mathcal{F}(W) \subset W$ (show that $W \neq 0$?).

2. Determine all eigenvalues of $\mathcal{F}$ as an $\mathbb{R}$ -linear transformation.
3. Determine all eigenvalues of $\mathcal{F}$ as a $\mathbb{C}$ -linear transformation.